

Natural Language Processing

Final Project

Abstractive Arabic Text Summarization Using Sequence-to-Sequence Models

1. Project Overview

Automatic text summarization is a core task in Natural Language Processing (NLP) that aims to generate a concise and meaningful summary of a given document while preserving its key information. There are two main paradigms:

- Extractive summarization, which selects important sentences or phrases from the original text, and
- Abstractive summarization, which generates new text that captures the essence of the source text in a more semantic form.

In this project, you will build an abstractive summarization system for Arabic text using neural Sequence-to-Sequence (Seq2Seq) learning (e.g., LSTM/GRU encoder-decoder) with an attention mechanism. Seq2Seq models learn how to map input sequences (long Arabic documents) to output sequences (summaries).

2. Objectives

By the end of this project, students should be able to:

1. Understand Seq2Seq architectures with encoder and decoder networks.
2. Implement attention mechanisms to improve abstraction quality.
3. Perform Arabic text preprocessing, including tokenization and normalization.
4. Train and evaluate a deep learning model for abstractive summarization.
5. Apply evaluation metrics such as ROUGE to assess summary quality.

3. Datasets

Students will start by loading Arabic summarization datasets that contain document-summary pairs. The Datasets you should use for this project:

1. **SumArabic** – Arabic abstractive summarization dataset with ~84,000 news article-summary pairs (train/validation/test splits):
[SumArabic dataset on Mendeley Data \(Arabic news\)](#)
2. **AraSum** – Large monolingual Arabic summarization corpus (~49,604 examples):
[AraSum dataset on GitHub \(JSON corpus\)](#)
3. **Egyptian Arabic Text Summarization Dataset** – Focused on Egyptian dialect text summarization, useful for dialectal evaluation:
[Egyptian Arabic Summarization dataset on Hugging Face](#)

4. Arabic Text Summarization:

<https://www.kaggle.com/datasets/abdalahmanshahrour/arabicsummarization>

Note: Some datasets may require citation or registration. Always verify licenses and usage terms.

4. Model Design

Students will implement a deep neural Seq2Seq model with Attention, using the following options:

- Attention-Based Seq2Seq
 - o Encoder: Bidirectional LSTM/GRU
 - o Decoder: LSTM/GRU + softmax output
 - o Bahdanau or Luong Attention mechanism to improve summary quality
- Design User interface App using Streamlit to facilitate using the Model.

5. Training and Evaluation

Training should be done on the training split of the dataset with early stopping on the validation set. Evaluation metrics typically include:

- ROUGE-1, ROUGE-2, ROUGE-L scores (higher means better overlap with real summaries)
- Qualitative analysis: sample summaries vs reference summaries

6. GitHub Repo

Every team should create a repo on Github and share it with you instructor for the sake of follow up. In this regards you should learn how to use git and bash commands.

7. Expected Deliverables

1. **Project Report** – Describing your approach, preprocessing, model architecture, training curves, evaluation metrics, and error analysis.
2. **Codebase** – Fully working Python code (TensorFlow or PyTorch) for the summarization pipeline. It must be uploaded on the Github.
3. **Results** – Plots of loss/ROUGE scores and example summaries compared with gold summaries.
4. **Presentation & Discussion** – Short presentation (slides) summarizing the work.

8. Grading Criteria

Component	Due Date	Weight
Data preprocessing	Week 5 (16/3/2026)	10%
Model design & implementation & Training & evaluation	Week 11 (29/4/2026)	40%
Report clarity & Results analysis	Week 12 (06/5/2026)	15%
Building User interface using Streamlit or Gradio	Week 13 (13/5/2026)	10%
Discussion	Week 14 (20/5/2026)	25%

9. Use of AI Tools

Students may collaborate with classmates and use Generative AI (Assistive Use) while studying the course, as long as they avoid direct plagiarism as outlined below. Students are allowed to seek assistance using artificial intelligence tools for supportive, non-essential purposes, such as brainstorming ideas, generating suggestions, proposing preliminary or general ideas or titles, linguistic proofreading, rephrasing and reformulation, analyzing results, and accessing sources and references. This does not include drawing conclusions. However, it is not allowed to copy and include the content directly in the final submitted work. The permitted percentage of AI-generated content (Assistive Use) must not exceed 40% of the total submitted academic, code or research content.

10. Plagiarism

Using a tool does not transfer authorship. If you use any tool to generate code, you are not considered the author of that code. You cannot submit code generated by another person/tool and present it as your own. This constitutes serious plagiarism and will be reported to the Dean.